

Ex.No:1: Simulation of FDMA in MATLAB

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

Objective 1:

To simulate an FDMA system by allocating **individual frequency channels** to multiple users and determine the **channel bandwidth** assigned to each user.

Output Parameter: Channel Bandwidth (kHz)

Procedure :

1. Open **MATLAB** and create a new script file.
2. Define the total available bandwidth, number of FDMA users, carrier frequencies, signal power, and noise power.
3. Allocate a unique frequency channel to each user and calculate the **channel bandwidth**.
4. Compute the **data rate** for each user based on the allocated bandwidth and modulation scheme.
5. Calculate the **Signal-to-Noise Ratio (SNR)** using the signal and noise power values.
6. Plot the transmitted signals of all FDMA users using different carrier frequencies.
7. Display the calculated **channel bandwidth, data rate, and SNR** in the MATLAB Command Window.
8. Analyze the results to verify independent channel allocation and evaluate the overall performance of the FDMA system.

Program (Matlab Code)

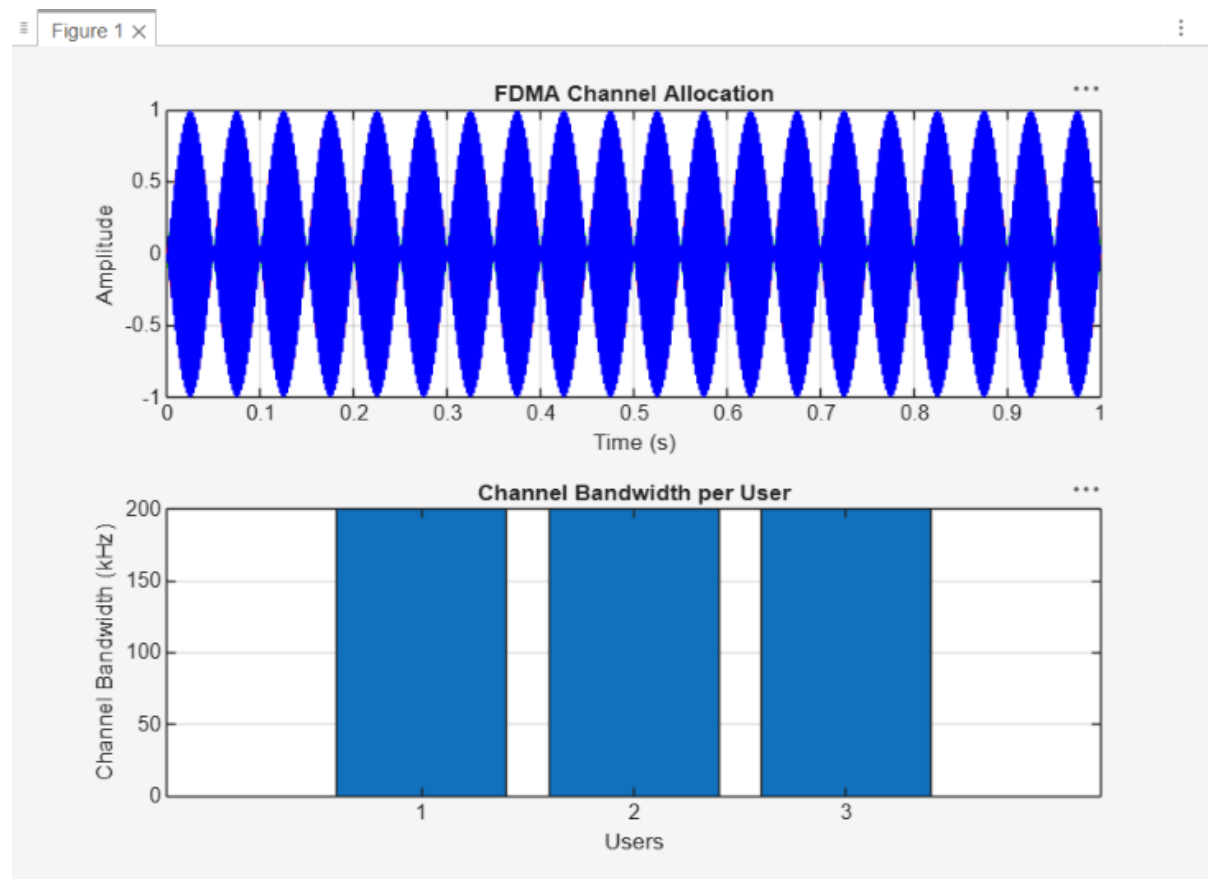
```
clc;
clear;
close all;
Fs = 1000;
time = 0:1/Fs:1;
totalBW = 600;
numUsers = 3;
channelBW = totalBW / numUsers;
carrier = [100 300 500];
msg = sin(2*pi*10*time);
user1 = msg .* cos(2*pi*carrier(1)*time);
user2 = msg .* cos(2*pi*carrier(2)*time);
user3 = msg .* cos(2*pi*carrier(3)*time);
```

```

subplot(2,1,1)
plot(time,user1,'r',time,user2,'g',time,user3,'b')
grid on
title('FDMA Channel Allocation')
xlabel('Time (s)')
ylabel('Amplitude')
subplot(2,1,2)
bw = channelBW * ones(1,numUsers);
bar(1:numUsers,bw)
grid on
title('Channel Bandwidth per User')
xlabel('Users')
ylabel('Channel Bandwidth (kHz)')

```

OUTPUT :



Result:

The FDMA system was successfully simulated by allocating separate frequency channels to three users, and the channel bandwidth assigned to each user was determined.

Objective 2:

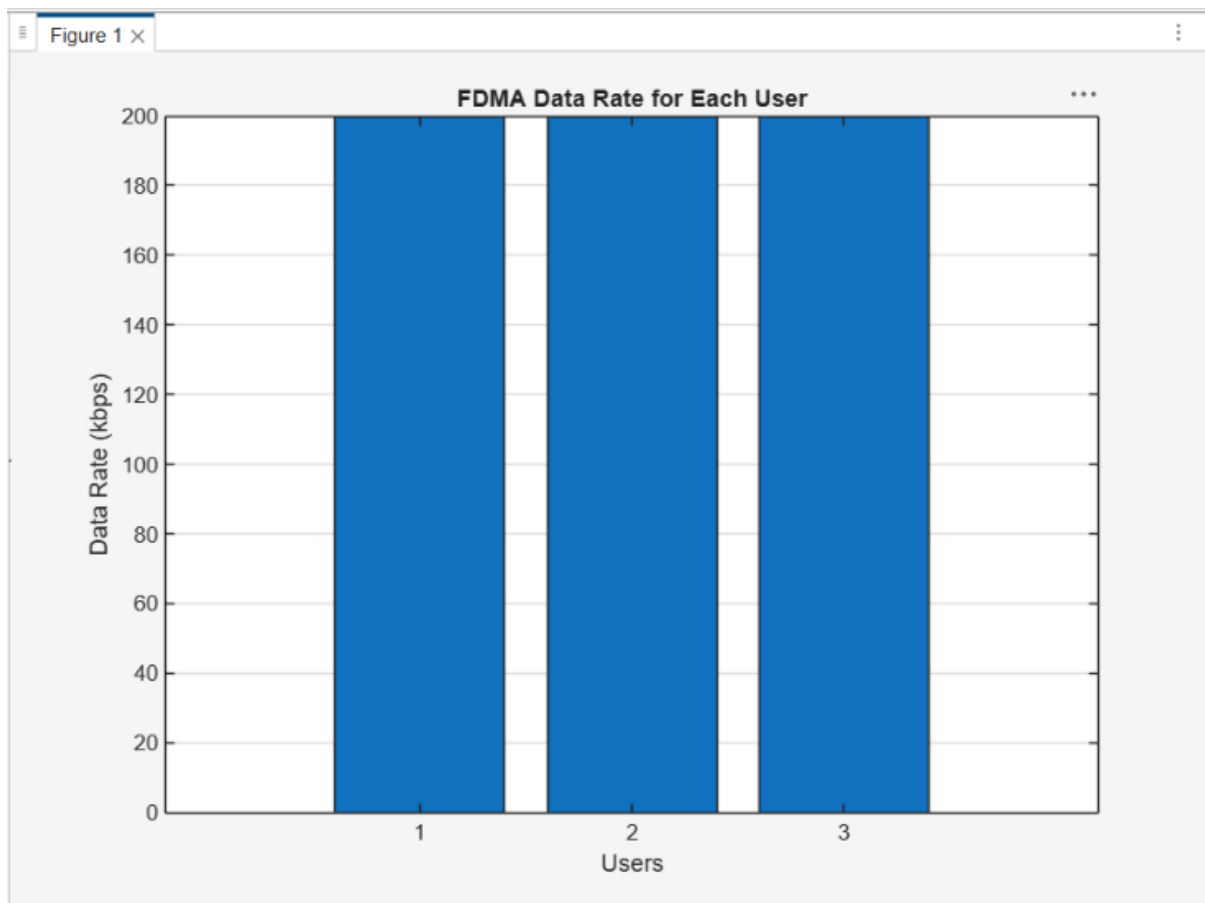
To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth.

Output Parameter: Data Rate (kbps)

PROGRAM :

```
clc;
clear;
close all;
channelBW = 200;
numUsers = 3;
modOrder = 2;
userRate = channelBW * log2(modOrder);
dataRate = repmat(userRate,1,numUsers);
disp('User Data Rate (kbps)')
disp([(1:numUsers)' dataRate'])
figure
bar(1:numUsers,dataRate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on
```

OUTPUT



RESULT :

The data rate achieved by each user was successfully calculated based on the allocated channel bandwidth, demonstrating equal transmission capacity for all users.

Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-toNoise Ratio (SNR) for different users.

Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

PROGRAM:

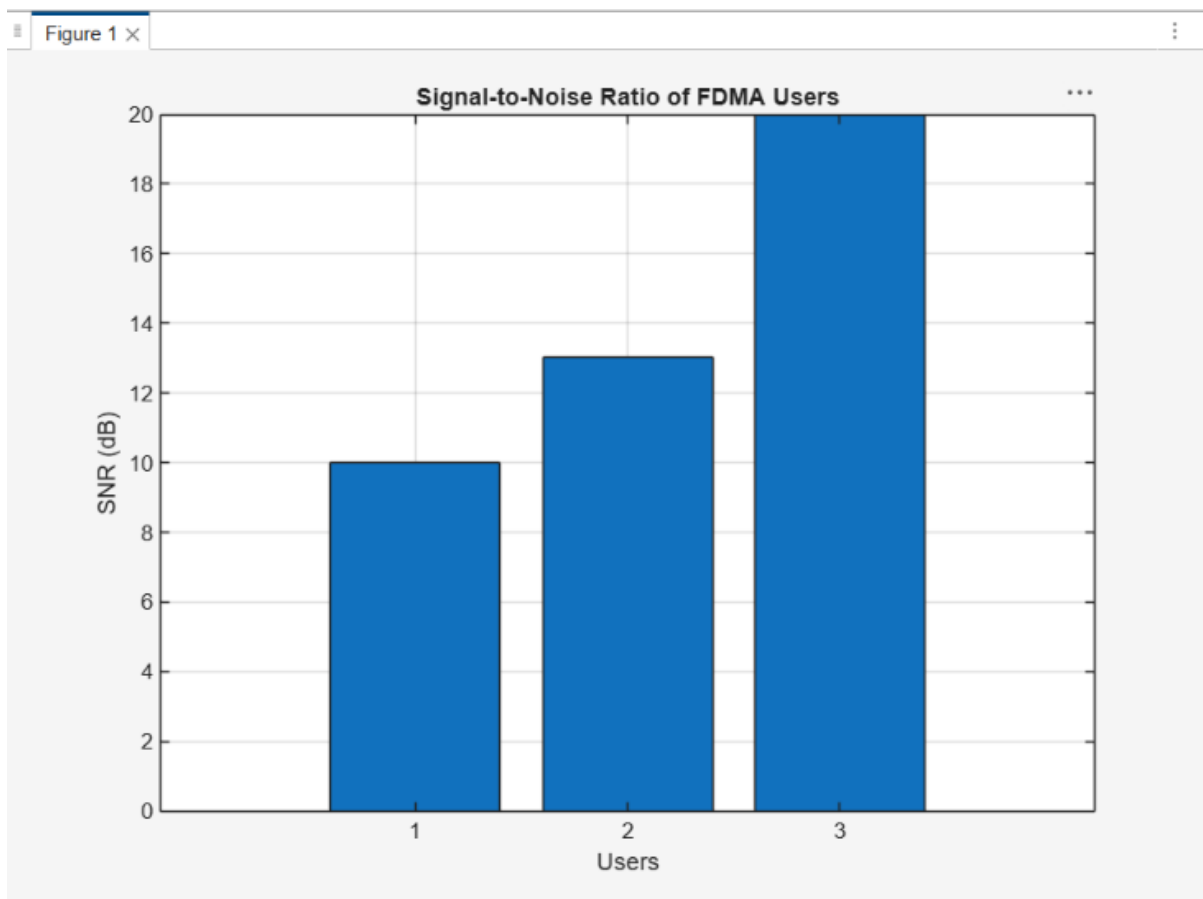
```
clc;
clear;
close all;
signalPower = 1;
noisePower = [0.1 0.05 0.01];
userID = 1:3;
```

```

snrValue = 10 * log10(signalPower ./ noisePower);
disp('User  Noise Power(W)  SNR(dB)')
disp([userID' noisePower' snrValue'])
figure
bar(userID,snrValue)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on

```

OUTPUT:



RESULT:

The Signal-to-Noise Ratio (SNR) for different users was successfully evaluated, showing that higher SNR values provide better communication quality and reliable data transmission.

CODE explanation

<code>clc;</code>	<code>% Clears the Command Window output</code>
<code>clear;</code>	<code>% Removes all variables from the MATLAB workspace</code>
<code>close all;</code>	<code>% Closes all open figure windows</code>
<code>channelBW = 200;</code>	<code>% Assigns the total channel bandwidth as 200 kbps</code>
<code>numUsers = 3;</code>	<code>% Specifies the total number of FDMA users</code>
<code>modOrder = 2;</code>	<code>% Sets the modulation order (2 = Binary modulation/BPSK)</code>
<code>userRate = channelBW * log2(modOrder);</code>	<code>% Calculates the data rate for one user</code>
<code>dataRate = repmat(userRate,1,numUsers);</code>	<code>% Creates a row vector by repeating the user data rate</code>
<code>disp('User Data Rate (kbps)')</code>	<code>% Displays the heading in the Command Window</code>
<code>disp([(1:numUsers)' dataRate'])</code>	<code>% Displays user number and corresponding data rate</code>
<code>Figure</code>	<code>% Opens a new figure window for plotting</code>
<code>bar(1:numUsers,dataRate)</code>	<code>% Creates a bar graph</code>
<code>xlabel('Users')</code>	<code>% Labels the X-axis as "Users"</code>
<code>ylabel('Data Rate (kbps)')</code>	<code>% Labels the Y-axis as "Data Rate (kbps)"</code>
<code>title('FDMA Data Rate for Each User')</code>	<code>% Adds the graph title</code>
<code>grid on</code>	<code>% Displays grid lines on the graph for better readability</code>